

MECHANICAL ENGINEERING MATERIALS*Time Allowed: 3 Hours**Full Marks: 60***Answer the following questions from Group-A, B & C as directed.****Group A**

1. Choose the correct answer from the alternatives (any ten): 1 x 10
- i. When a steel is heated in a furnace and then cooled in air at ordinary temperature, the process is
(a) Annealing (b) Hardening (c) Normalizing (d) Tempering
 - ii. γ form of iron is?
a) Magnetic b) Non magnetic c) Occurs below 768°C d) Occurs above 1410°C
 - iii. Bronze is an alloy of
(a) copper & zinc (b) copper & tin (c) copper, zinc & tin (d) any one of these
 - iv. Fatigue failure can be detected with the help of – a) stress – strain curve, b) S – N curve, c) graph of law of machine, d) iron – carbon diagram.
 - v. In low carbon steel C % varies from- (a) 0.008-0.3 (b) 0.3-0.6 (c) 0.6-2.0 (d) 2.0-4.3
 - vi. In the austempering process of heat treatment, austenite changes into
a) martensite b) troostite c) sorbite d) bainite
 - vii. Gold has the crystal structure of- (a) BCC (b) FCC (c) HCP (d) RCC
 - viii. The name diffusion less transformation is associated with –
a) pearlite, b) bainite, c) martensite, d) cementite.
 - ix. Silicon when added to copper improves-
(a) machinability, (b) hardness, (c) hardness and strength, (d) strength and ductility
 - x. The ability of a material to be drawn into thin wire is known as-
a) Malleability b) Ductility c) Harness d) Brittleness
 - xi. The temperature (in °C) of peritectic reaction in iron-carbon phase diagram is-
a) 1539 b) 1493 c) 910 d) 1175
 - xii. The machine tool guide ways are usually hardened by
(a) vacuum hardening, (b) martempering, (c) induction hardening, (d) flame hardening
 - xiii. The hardenability is not affected by -
(a) air, (b) chemical composition of steel, (c) critical cooling rate, (d) quenching medium and method of quenching
 - xiv. To obtain martensite structure – a) slow cooling is to be done, b) quick cooling is to be done, c) air cooling is to be done, d) the cooling curve must cross C- curve.
 - xv. Stainless steel is so called because of its-
(a) High strength (b) High corrosion resistance
(c) High ductility (d) Brittleness

2. Fill in the blanks (any ten):

1x10

- a) The number of effective atoms in a HCP unit cell is _____.
- b) The ultimate stress of a material is _____ than the breaking stress.
- c) Plastic cables are made by _____ process.
- d) _____ metal has the highest electrical conductivity.
- e) Curie temperature for iron is _____.
- f) The iron ore mostly used for the production of pig iron is _____.
- g) Carbon percentage in cast iron is _____.
- h) Most metals has _____ structure.
- i) Y-alloy is an important alloy of _____.
- j) The steel used for making shafts is _____.
- k) Babbit metal is a _____ based alloy.
- l) _____ as a medium is used for fastest cooling rate of steel quenching.
- m) The cupola is used to manufacture _____.
- n) BUNA- N is also called _____.
- o) The heat treatment process used for softening of hardened steel is _____.

3. Answer the following questions (any ten): <https://www.wbscteonline.com>

1x10

- a) Why polymer cannot be remoulded?
- b) What is the full form of PVC?
- c) What is creep?
- d) What is the last step in magnetic particle test method?
- e) What is 18:4:1 tool steel?
- f) Give two examples of Thermoplastics.
- g) Write two applications of Muntz metal.
- h) Give two examples of non destructive testing.
- i) Briefly state the application of Inver.
- j) What is the unit of strain?
- k) What is unit cell?
- l) What is meant by work hardening?
- m) What is the difference between bronze and brass?
- n) What does ASTM stand for?
- o) What is radiography test?

Group B

4. Answer the following questions (any six):

2x6

- i. What are the desirable properties of bearing materials?
- ii. How plain carbon steel is classified?
- iii. What are composite materials?
- iv. Write down the difference between hardness & toughness.
- v. What is Duralumin?
- vi. Write the composition of '40 Ni 2 Cr 1 Mo 25' steel.
- vii. What is an 'Allotropic metal'?
- viii. What is powder metallurgy process?
- ix. What is the purpose of annealing?
- x. What is phase diagram?

Group C

5. Answer the following question (any one):

1x6

- a) Draw the 'Iron-carbon equilibrium diagram' and explain it. Briefly describe Surface hardening and tempering as heat treatment processes.

4+2

- b) Determine the average number of atoms, relation between lattice parameter and atomic radius, atomic packing factor of BCC crystal structure. 2+2+2
- c) What do you mean by a unit cell? Volume of a unit cell and one atom of the unit cell are 100 nm^3 and 10 nm^3 respectively. If the unit cell has 5 numbers of effective atoms then find out the atomic packing factor (APF) of the unit cell. 2+4
6. Answer the following question (any one): 1x6
- a) Explain the necessity of heat treatment for steel. Describe the process of quenching. 4+2
- b) Write short notes on - (i) induction hardening, (ii) flame hardening 3+3
- c) Describe normalizing process. State its purpose. 4+2
7. Answer the following question (any one): 1x6
- a) State two differences between thermosetting plastics and thermoplastics with proper examples. Give one application for each of the followings- (a) ABS, (b) Acrylics, (c) Bakelite. 3+3
- b) Differentiate between destructive testing and non destructive testing. Write short note on – i) Dry penetration test, ii) Magnaflux test. 2+2+2
- c) What is a polymer ? Give two examples. What is Vulcanized Rubber? 2+2+2

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